



Department of Information Technology

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch : V Semester-B. Tech Information Technology

Course Code & Title : IT3501 Full Stack Web Development

Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 3 / Managing Collections
- **Course Outcome** : CO3
- **Topic Learning Outcome:** TLO13
- **Activity Chosen** : Learning by Teaching
- **Justification:**
 - Learning by Teaching is implemented as an innovative instructional strategy to teach the topic Managing Collections in MongoDB, allowing students to act as instructors and explain how collections are created, structured, and handled within a database to their peers.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
 - Learning by Teaching involves students taking responsibility for learning a concept thoroughly and then teaching it to others, thereby reinforcing their own understanding.
 - One week prior to the class, Ms. M. Valli was allotted the topic Managing Collections in MongoDB to ensure she had enough time to organize her content.
 - During the session, Ms. M. Valli illustrated the essential steps involved in creating, modifying, and handling MongoDB collections, using clear examples on the board to help her peers visualize the database operations.
 - The student-led presentation spanned around 30 minutes, during which various subtopics including creating a collection, inserting documents, updating and deleting records, and viewing collection structures were explained.
 - After the presentation, queries related to the topic were addressed, and the concepts were consolidated with a short summary.
 - The classroom implementation of this Learning by Teaching activity is shown in Figure1.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO3	3	2	2	2	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	2	2	2	2	3
Justification for correlation	The Students will be able to gain a clear understanding of the operations involved in creating and managing collections in MongoDB.	The Students will be capable of performing collection-related tasks such as insert, update, and delete operations using MongoDB commands or GUI tools.	Students enhance their ability to design and organize database structures effectively.	Taking responsibility for teaching boosts the student instructor's individual confidence and strengthens their mastery of MongoDB fundamentals.	The Students presentation skill will be improved as they are facing the audience and teaching the concepts	Students become competent in applying collection-handling concepts while developing MongoDB-based applications.

- **Images / Screenshot of the practice:**



Figure 1: Screenshot of Ms.M.Valli Learning by teaching activity presentation

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students listened attentively and expressed that they clearly understood the procedures for creating and managing collections in MongoDB.
- By presenting the topic to their peers, students improved their confidence levels and gained detailed insight into handling collections effectively in MongoDB.

- ❖ ***Benefit of the practice:***

- The activity strengthened students' understanding of managing collections in MongoDB, a core skill required for organizing, storing, and handling data effectively in modern database-driven applications.
- This activity enhanced students' technical understanding and improved their communication skills through active peer interaction.
- Students were able to share insights and practical experiences related to handling collections in MongoDB.

- ❖ ***Challenges faced in implementation:***

- More time was needed for students to fully share their thoughts and perspectives.
- A few students lacked basic knowledge of MongoDB, which made collection management difficult for them.

At the end of session, the instructor restated the basics of MongoDB collection management. Students were shown the proper syntax to correct mistakes in creating collections and adding documents.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1_VA_CS8603_Learning_teaching.pdf
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/6_GS_CS8592_learning_by_teaching.pdf

Signature of Faculty Member

HOD